

Ticagrelor & Prasugrel — Reversible vs Irreversible P2Y₁₂

ANTIPLATELET SCENE 5 — “THE SPRING-LATCH PLAYER”

Ticagrelor - Brilinta

ZONE 1 — LEFT: MECHANISM + DOSING + INDICATIONS

ZONE 2 — CENTER: CYP3A4 INTERACTIONS + RENAL/HEPATIC

ZONE 3 — RIGHT: SIDE EFFECTS

ZONE 4 — FAR RIGHT: REVERSAL + BENTRACIMAB

INDICATIONS

- ACS — BOTH medical management AND stent
- SECONDARY PREVENTION — up to 3 YEARS in prior MI patients at high risk
- PREFERRED OVER CLOPIDOGREL FOR ACS + PCI

ASPIRIN ≤100mg/day ONLY
 300mg loading OK once ✓
 →100mg daily ✗ — REDUCES TICAGRELOR EFFICACY

NOT A HOT DOG

CYP3A4 *metabolize*

1 P2Y₁₂
 SPRING LATCH bot complex locks clips on, but can pop open and release at any time

2
 180mg (LOADING)
 spring can pops to gate reopens

3
 90mg BID

7 CYP3A4 INHIBITORS (TOXIC)
 CYP3A4 INHIBITORS CAOKING MACHINE - DRUG OVERHEATES. PILUG UP EVERYWHERE → DRUG ACCUMULATION AND TOXIC LEVELS WITH MASSIVE BLEEDING RISK

8 CYP3A4 INDUCERS (NO PROTECTION)
 CYP3A4 INDUCERS (NO PROTECTION) MAKING CYP3A4 MACHINE WORKS. - BURNING THROUGH TICAGRELOR INSTANTLY - SUBTHERAPEUTIC LEVELS - NO PLATELET PROTECTION. RISKING AN MI

RENAL/HEPATIC

KIDN NO DOSE ADJUSTMENT IN RENAL INSUFFICIENCY

LIVER CAUTION — CYP3A4 metabolized; impaired liver → unpredictable drug levels

11 VENTRICULAR PAUSES
 ASYMPTOMATIC VENTRICULAR PAUSES
 ASYMPTOMATIC VENTRICULAR PAUSES
 MONITOR IN PATIENTS WITH PRE-EXISTING BRADYCARDIA

12 BLEEDING
 STOP ≥5 DAYS BEFORE MAJOR SURGERY.

COMPARISON	OVERALL
MAJOR BLEEDING (ICH, major GI)	NO SIGNIFICANT DIFFERENCE vs clopidogrel
MINOR BLEEDING	MORE FREQUENT with ticagrelor
OVERALL	MORE BLEEDING with ticagrelor

13 STROKE HISTORY BONUS
 Like clopidogrel
 NO CONTRAINDICATION AFTER STROKE

15 BENTRACIMAB
 HUMANIZED ANTIBODY FRAGMENT (Fab)
 BINDS TICAGRELOR + ITS ACTIVE METABOLITE WITH HIGH AFFINITY
 Leaving platelets' P2Y₁₂ gates free open

15 Future reversal agent — Fab fragment, binds ticagrelor + STATUS: UNDER DEVELOPMENT — NOT YET STANDARD OF CARE

15 CONTRAST: PRASUGREL PRASUGREL IS ABSOLUTELY CONTRAINDICATED IN PRIOR STROKE like clopidogrel

15 POST-STROKE + ASPIRIN 2-3d.

15 PLATELET TRANSFUSION UNLIKELY TO HELP — FREE DRUG REBINDS NEW PLATELETS

15 PLATELETMOGRELOR JUMPE O FRE FREE DONOR P2Y₁₂ GATES

15 TWICE DAILY — clopidogrel and prasugrel are once daily!

15 ASPIRIN ≤100mg ONLY, >100mg reduces efficacy

15 ACS — medical + stent, 3-year secondary prevention, preferred over clopidogrel

15 NOT a prodrug, no hepatic activation, CYP2C19 irrelevant

15 Spring latch on P2Y₁₂ gate: reversible binding

15 180mg loading, 90mg BID: dosing — note twice daily (unique in class)

15 Aspirin co-administration rule

1. Ticagrelor banner

WHAT YOU SEE

Title 'ANTIPLATELET SCENE 5 — THE SPRING-LATCH PLAYER' with 'Ticagrelor / Brilinta' written in red at top right.

WHAT IT ENCODES

Ticagrelor (brand Brilinta) is a reversible, direct-acting P2Y12 ADP-receptor antagonist. Unlike the thienopyridines it is NOT a prodrug, so it needs no hepatic CYP2C19 activation and gives faster, more consistent platelet inhibition. Dosed 180 mg load then 90 mg BID.

BOARD TRAP

Don't call ticagrelor a thienopyridine or prodrug — it is a cyclopentyltriazolopyrimidine that binds reversibly and works directly without CYP2C19 activation.

2. Aspirin dose ceiling

WHAT YOU SEE

Red handwriting: ASPIRIN >100 mg daily / >100 mg loading lowers efficacy → reduce TICAGRELOR efficacy.

WHAT IT ENCODES

Ticagrelor must be paired with LOW-dose aspirin (<= 100 mg/day maintenance). Higher aspirin doses paradoxically reduce ticagrelor's benefit (seen in PLATO subgroup). This is a unique ticagrelor-specific interaction.

BOARD TRAP

High-dose aspirin (>100 mg) lowers ticagrelor efficacy — a trap unique to ticagrelor; standard for clopidogrel/prasugrel it does not apply.

3. 180 mg loading egg

WHAT YOU SEE

Egg labeled '180 mg LOADING' on the grill.

WHAT IT ENCODES

Ticagrelor loading dose is 180 mg orally, then 90 mg twice daily maintenance. Rapid onset because no hepatic activation step is required.

BOARD TRAP

180 mg load / 90 mg BID is ticagrelor — do not confuse with prasugrel's 60 mg load / 10 mg once-daily or clopidogrel's 300–600 mg load / 75 mg daily.

4. P2Y12 gate — reversible binding

WHAT YOU SEE

Spring/latch 'P2Y12' gate that pops open again; note 'reversible binding'.

WHAT IT ENCODES

Ticagrelor binds the P2Y12 receptor REVERSIBLY at an allosteric site, so platelet function recovers as drug clears (offset ~3–5 days). Irreversible binders (clopidogrel, prasugrel) need new platelets to regenerate function (~7–10 days).

BOARD TRAP

Reversible offset means ticagrelor washes out faster than prasugrel/clopidogrel — but BID dosing is mandatory because of that same reversibility.

5. 90 mg BID egg

WHAT YOU SEE

Egg labeled '90 mg BID' in the center.

WHAT IT ENCODES

Maintenance ticagrelor is 90 mg twice daily (BID). The twice-daily requirement is a direct consequence of reversible, short-acting binding — missed doses quickly restore platelet activity.

BOARD TRAP

Twice-daily = ticagrelor. A once-daily P2Y12 inhibitor is prasugrel (10 mg) or clopidogrel (75 mg); BID is the ticagrelor fingerprint.

6. CYP3A4 liver — interactions

WHAT YOU SEE

Liver labeled 'CYP3A4' in the center 'CYP3A4 INTERACTIONS' zone.

WHAT IT ENCODES

Ticagrelor is metabolized by CYP3A4. Strong CYP3A4 inhibitors (ketoconazole, ritonavir, clarithromycin) raise levels and bleeding risk; strong inducers (rifampin, phenytoin, carbamazepine) lower levels and lose protection.

BOARD TRAP

Ticagrelor's key CYP is 3A4 (not 2C19). Clopidogrel/prasugrel depend on CYP2C19/3A4 for ACTIVATION; ticagrelor is already active and 3A4 just clears it.

7. CYP3A4 inhibitors (toxic)

WHAT YOU SEE

Mushrooms / 'CYP3A4 INHIBITORS (TOXIC)' with massive bleeding risk note.

WHAT IT ENCODES

Strong CYP3A4 inhibitors increase ticagrelor exposure → higher bleeding risk. Examples: azole antifungals, protease inhibitors, macrolide antibiotics. Avoid co-administration.

BOARD TRAP

A CYP3A4 inhibitor raises ticagrelor (more bleeding); an inducer lowers it (less protection) — opposite directions, commonly swapped on exams.

8. CYP3A4 inducers (no protection)

WHAT YOU SEE

'CYP3A4 INDUCERS (NO PROTECTION)' — subtherapeutic, thrombosis rising.

WHAT IT ENCODES

Strong CYP3A4 inducers (rifampin, phenytoin, carbamazepine, St John's wort) drop ticagrelor levels to subtherapeutic, risking stent thrombosis. Avoid in patients on ticagrelor.

BOARD TRAP

Inducer = loss of antiplatelet protection (thrombosis), not bleeding — easy to reverse the consequence.

9. Dyspnea 15%

WHAT YOU SEE

Big '15%' with 'DYSPNEA' as most common non-bleeding side effect; 'not responsive to bronchodilators'.

WHAT IT ENCODES

Dyspnea occurs in ~15% of ticagrelor patients — a hallmark adverse effect from adenosine accumulation, not bronchospasm, so it does NOT respond to bronchodilators and is usually benign/self-limited.

BOARD TRAP

Ticagrelor dyspnea is adenosine-mediated and bronchodilator-UNresponsive — don't treat as asthma/COPD or stop the drug reflexively.

10. Ventricular pauses

WHAT YOU SEE

ECG strip 'VENTRICULAR PAUSES' — asymptomatic, monitor.

WHAT IT ENCODES

Ticagrelor (via adenosine) can cause asymptomatic ventricular pauses/bradyarrhythmia, especially early. Usually benign; caution in sick-sinus or high-grade AV block without a pacemaker.

BOARD TRAP

Pauses and dyspnea share one mechanism — adenosine elevation — a ticagrelor-specific cluster absent with clopidogrel/prasugrel.

11. Stop 5 days before surgery

WHAT YOU SEE

'BLEEDING — stop 5 days before major surgery'.

WHAT IT ENCODES

Hold ticagrelor 5 days before major surgery (vs ~7 days for clopidogrel, ~7 days for prasugrel) because reversible binding gives a slightly shorter offset.

BOARD TRAP

Ticagrelor 5 days, clopidogrel 5–7 days, prasugrel 7 days pre-op — prasugrel needs the longest hold due to potent irreversible inhibition.

12. Stroke history bonus (prasugrel)

WHAT YOU SEE

'STROKE HISTORY BONUS' — contraindication note linked to prasugrel.

WHAT IT ENCODES

Prasugrel is CONTRAINDICATED in prior stroke or TIA (excess intracranial bleeding in TRITON-TIMI 38), and not recommended in age ≥ 75 or weight < 60 kg. Ticagrelor is the preferred potent agent in those patients.

BOARD TRAP

Prior stroke/TIA = avoid prasugrel, choose ticagrelor — this is the single most-tested prasugrel contraindication.

13. Prasugrel 60/10 mg

WHAT YOU SEE

Red note: 'CONTRAINDIC. PRASUGREL — POST-STROKE' near the reversal zone, and prasugrel dosing implied.

WHAT IT ENCODES

Prasugrel is a thienopyridine prodrug, irreversible P2Y₁₂ blocker, loaded 60 mg then 10 mg once daily. Most potent thienopyridine; reserved for ACS with PCI in patients without stroke history.

BOARD TRAP

Prasugrel is given AFTER coronary anatomy is known (at PCI), unlike ticagrelor/clopidogrel which can be pre-loaded upstream.

14. Bentracimab reversal

WHAT YOU SEE

Far-right 'REVERSAL + BENTRACIMAB' zone with antibody fragment and IV bag.

WHAT IT ENCODES

Bentracimab (PB2452) is an investigational humanized monoclonal antibody Fab fragment that binds and neutralizes ticagrelor, rapidly restoring platelet function. It is reversal specific to ticagrelor — under development, not yet standard of care.

BOARD TRAP

Bentracimab reverses ticagrelor only (it binds the free drug). It does NOT reverse clopidogrel or prasugrel, which are irreversible and need platelet transfusion.

15. Platelet transfusion futility

WHAT YOU SEE

Blood bag 'PLATELET TRANSFUSION' note — clopidogrel/prasugrel jump fresh platelet gates.

WHAT IT ENCODES

For irreversible thienopyridines (clopidogrel, prasugrel) the only rescue for bleeding is platelet transfusion. For ticagrelor, transfused platelets are quickly inhibited by circulating drug, so transfusion is far less effective — hence bentracimab.

BOARD TRAP

Platelet transfusion helps clopidogrel/prasugrel bleeds but is largely futile for ticagrelor — its reversible drug keeps binding new platelets.